# Data Science 500 Assignment

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| QUESTION | EXAMINER MARKS | MODERATOR MARKS | REMARKS |
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## ASSIGNMENT INSTRUCTIONS

*Please tick each box to confirm completion.*

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

**For essay-style assignments, please include the following sections:**

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

**DECLARATION OF ORIGINALITY:**

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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## 1. Introduction

This project analyses book listings from Books to Scrape, a practice website designed for web-scraping exercises. The objectives are to collect book information, prepare a structured dataset, explore price and rating patterns, and build a linear regression model to estimate prices. The supplied cleaned dataset contains 200 unique book records (Books to Scrape, 2017).

## 2. Methodology

Python, Requests, Beautiful Soup, pandas, Matplotlib, and scikit-learn are used in the workflow. The notebook requests catalogue pages and parses each book card with Beautiful Soup. The extracted fields include book title, price, star rating, stock availability, and quantity where the listing provides it. The notebook is configured to scrape the first ten catalogue pages, which normally provides approximately 200 listings. HTTP errors are checked and a short pause is included between page requests.

Prices are converted to numeric GBP values. Star ratings are mapped from the words One to Five to numeric values 1–5, while the word rating is retained. Duplicate titles and rows without a usable title, price, or rating are removed (Beautiful Soup, 2006).

## 3. Data Cleaning and Preparation

The cleaned dataset contains 200 records and 7 columns. The average listed price is £34.80, the median is £35.64, and the observed price range is £10.16 to £59.64. The average numeric rating is 2.87 out of 5.

Stock quantity is missing for 200 of the 200 records. This is treated as unknown rather than replaced with zero, because a numeric quantity is not present in the supplied listing dataset. All 200 books are marked as in stock, so this sample does not provide variation in stock status and cannot support conclusions comparing in-stock with out-of-stock books.

## 4. Data Visualisation and Exploration

The four visualisations examine the price distribution, rating counts, the relationship between rating and price, and average price by rating. Prices range from £10.16 to £59.64; the mean is £34.80 and the median is £35.64.

Rating counts are: one star, 49; two stars, 41; three stars, 35; four stars, 38; and five stars, 37. The average prices for ratings of one to five stars are £34.59, £33.74, £36.19, £34.90, £34.82, respectively. These averages

are relatively similar. The correlation between rating and price is 0.017, indicating almost no linear relationship in this sample. Star rating alone therefore does not appear to be a strong indicator of price. These descriptive findings apply to this 200-book sample and should not automatically be generalised to the entire book market (scikit-learn,2008).

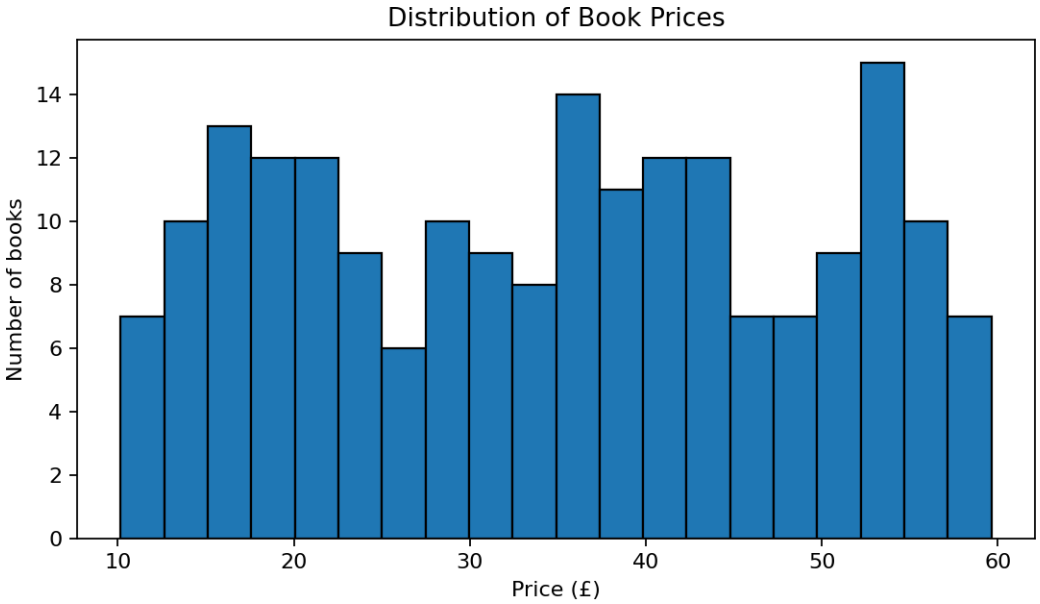



Figure 1. Distribution of book prices.

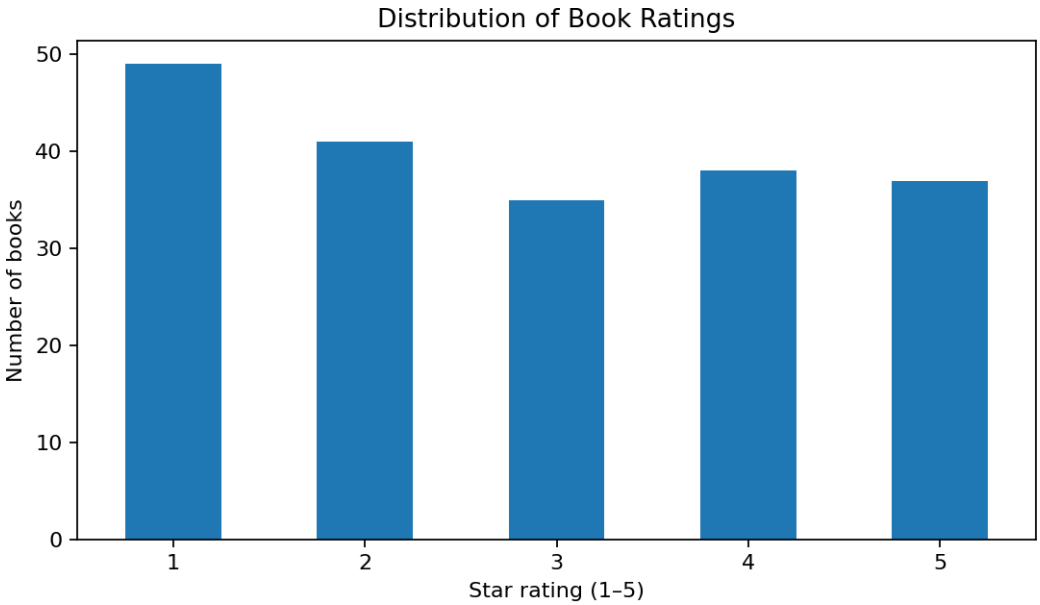


Figure 2. Counts of books by star rating.

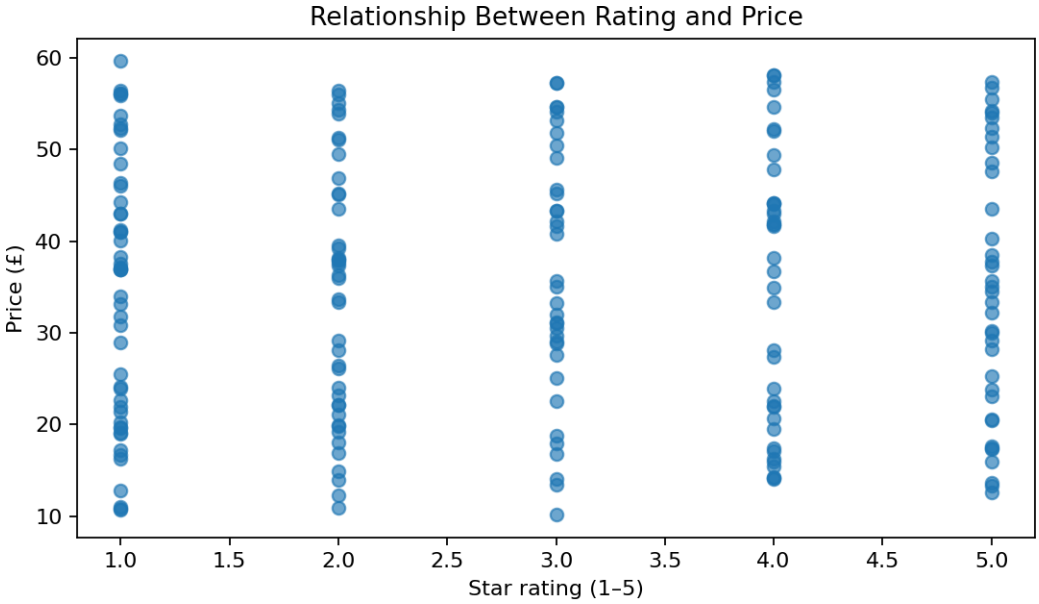



Figure 3. Relationship between star rating and price.




Figure 4. Average price by star rating.

## 5. Predictive Modelling

A linear regression pipeline estimates price from two predictors: book title represented as TF-IDF text features and numeric star rating. The data is split into training and test sets using an 80/20 split and a fixed random state of 42 so the split can be reproduced. The model is evaluated on the held-out test set.

The saved notebook output uses 160 training rows and 40 test rows. Its metrics are MAE = £18.84, MSE = £537.07 GBP², RMSE = £23.17, and R-squared = -1.327. These values are aligned with the notebook output.

| Metric | Result |
|---|---|
| Training rows | 160 |
| Test rows | 40 |
| MAE | £18.84 |
| MSE | £537.07 GBP² |
| RMSE | £23.17 |
| R-squared | -1.327 |

MAE is the average absolute prediction error in pounds. MSE is the average squared prediction error, which penalises larger errors. RMSE expresses prediction error in pounds. Negative R-squared means the model performed worse than a baseline that predicts the test-set mean on this test split. The title and rating features are insufficient to predict prices accurately in this sample (scikit-learn, 2008).

| Metric | Linear regression | Mean-price baseline |
|---|---|---|
| MAE | £18.84 | £13.40 |
| RMSE | £23.17 | £15.24 |
| R-squared | -1.327 | -0.007 |

## 6. Discussion and Recommendations

The dataset demonstrates an end-to-end workflow, but it does not establish strong pricing rules. The average prices for different ratings are similar, and the rating-price correlation is close to zero. The model's negative R-squared reinforces that the selected features do not reliably explain price variation in the held-out sample. The title may be unique rather than predictive of prices for other books. Missing stock quantities also prevent analysis of numeric stock levels.

Recommendations:

Scrape more catalogue pages or the full catalogue to increase the number and variety of records.

Where available, collect genre/category and other relevant book attributes, then test whether they improve predictions.

Keep unknown stock quantities missing unless a source explicitly supplies a numeric quantity.

Compare the linear regression model with a simple mean-price baseline and validate using held-out data.

Re-run the notebook before submission if live scraping succeeds and update the report metrics if the refreshed data differs.

## 7. Conclusion

The project demonstrates data acquisition with Beautiful Soup, cleaning of numeric and categorical fields, visual exploration, and evaluation of a linear regression model. In this 200-book sample, star rating has almost no linear relationship with price, and the baseline model does not predict prices accurately. The most important improvements are to expand the dataset, collect informative attributes, and keep data-quality decisions transparent.

## Bibliography

Beautiful Soup, 2006. Beautiful Soup Documentation. https://www.crummy.com/software/BeautifulSoup/bs4/doc/ [Accessed: 9 October 2026].

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scikit-learn, 2008 User Guide. https://scikit-learn.org/stable/user_guide.html [Accessed: 9 October 2026].